

Computer Networks (CS-241)

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Adapted From:

Computer Networking: A Top-Down Approach

8th edition: Jim Kurose, Keith Ross Pearson, 2020

Processes communicating

process: program running within a host

- When two or more processes exchange data or messages to perform a task, this is called process communication.

clients, servers

client process: process that initiates communication

server process: process that waits to be contacted

Types of Process Communication

- **Inter-Process Communication (IPC) – Same Machine**
- Used when processes run on the **same system**.

- ***Example:***

A text editor process sending data to a spell-checking process on the same computer.

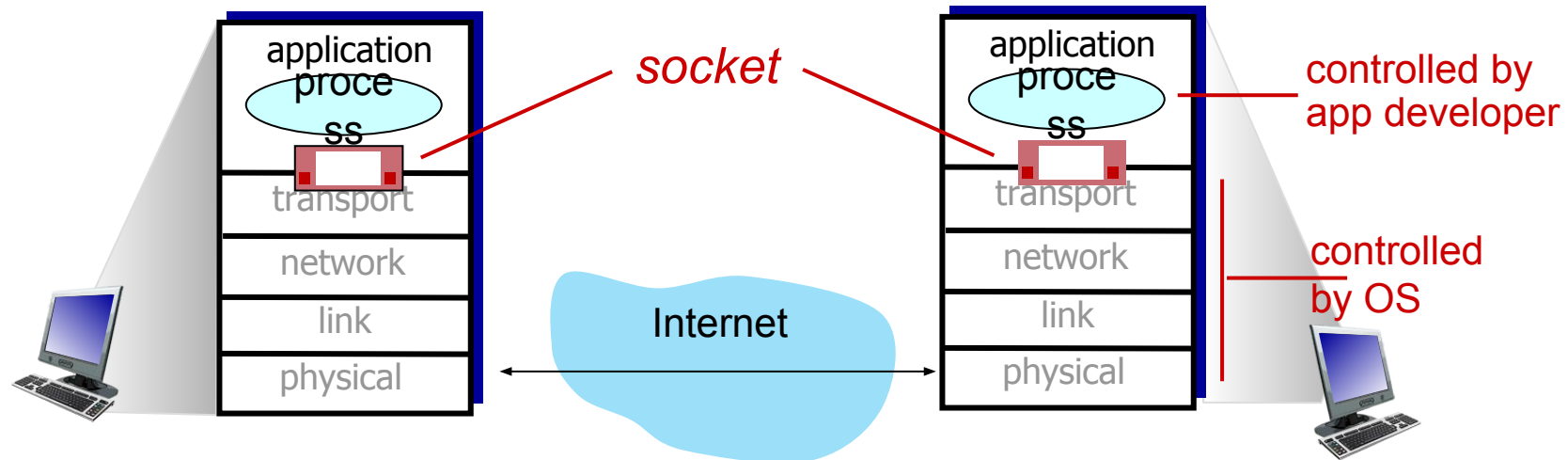
Types of Process Communication

Network Process Communication – Across different Machines

- Happens over a **network** (like the Internet).
 - Processes use **sockets** to send and receive data through **ports**.
 - Relies on **client–server architecture** or **peer-to-peer** models.
- *Example:*
When you use **WhatsApp**, your phone's app (client process) communicates with WhatsApp's **server process** to send or receive messages.

Sockets

- A socket is like a communication endpoint — it allows two processes (programs) to send and receive data across a network. Think of it as a door through which data enters or leaves your computer during communication.



Sockets

- Technical Definition:
- A **socket** is an **interface** provided by the Operating System that enables communication between:
- Processes on the same system (using **Inter-Process Communication**), or
- Processes on different systems connected over a **network** (using IP and ports).

A socket combines:

- IP address + Port number
- to uniquely identify a communication endpoint.

Example:

- 192.168.1.10 → IP address of the web server
- 80 → Port number for HTTP service
- So 192.168.1.10:80 → A socket for a web server

Real-Life Examples of Sockets

Application	Port	Protocol
HTTP (Web)	80	TCP
HTTPS (Secure Web)	443	TCP
DNS	53	UDP
FTP	21	TCP
WhatsApp	Dynamic ports	TCP/UDP

An application-layer protocol defines:

- types of messages exchanged,
 - e.g., request, response
- message syntax:
 - what fields in messages & how fields are described
- message semantics
 - meaning of information in fields
- rules for when and how processes send & respond to messages

open protocols:

- defined in RFCs, everyone has access to protocol definition
- allows for interoperability
- e.g., HTTP, SMTP

proprietary protocols:

- e.g., Skype, Zoom

Application layer: overview

- Principles of network applications
- **Web and HTTP**
- E-mail, SMTP, IMAP
- The Domain Name System DNS
- P2P applications
- video streaming and content distribution networks
- socket programming with UDP and TCP



Web and HTTP

First, a quick review...

- web page consists of *objects*, each of which can be stored on different Web servers
- object can be HTML file, JPEG image, Java applet, audio file,...
- web page consists of *base HTML-file* which includes *several referenced objects, each* addressable by a *URL*, e.g.,

`www.someschool.edu/someDept/pic.gif`

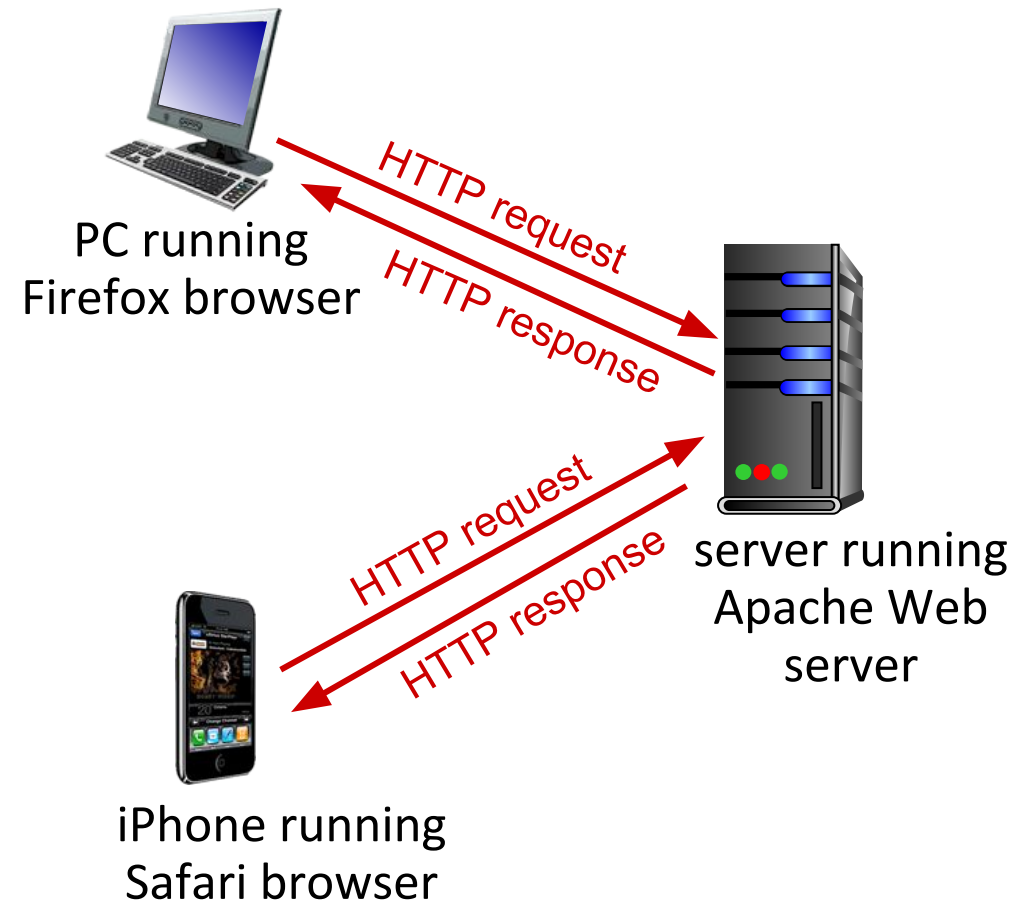
host name

path name

HTTP overview

HTTP: hypertext transfer protocol

- Web's application layer protocol
- client/server model:
 - *client*: browser that requests, receives, (using HTTP protocol) and “displays” Web objects
 - *server*: Web server sends (using HTTP protocol) objects in response to requests



HTTP overview (continued)

HTTP uses TCP:

- client initiates TCP connection (creates socket) to server, port 80
- server accepts TCP connection from client
- HTTP messages (application-layer protocol messages) exchanged between browser (HTTP client) and Web server (HTTP server)
- TCP connection closed

HTTP is “stateless”

- server maintains *no* information about past client requests

aside

protocols that maintain “state” are complex!

- past history (state) must be maintained
- if server/client crashes, their views of “state” may be inconsistent, must be reconciled

HTTP connections: two types

Non-persistent HTTP

1. TCP connection opened
 2. at most one object sent over TCP connection
 3. TCP connection closed
- downloading multiple objects required multiple connections

Persistent HTTP

- TCP connection opened to a server
- multiple objects can be sent over *single* TCP connection between client, and that server
- TCP connection closed

Non-persistent HTTP: example

User enters URL: `www.someSchool.edu/someDepartment/home.index`
(containing text, references to 10 jpeg images)



1a. HTTP client initiates TCP connection to HTTP server (process) at `www.someSchool.edu` on port 80



1b. HTTP server at host `www.someSchool.edu` waiting for TCP connection at port 80 “accepts” connection, notifying client

2. HTTP client sends HTTP *request message* (containing URL) into TCP connection socket. Message indicates that client wants object `someDepartment/home.index`

3. HTTP server receives request message, forms *response message* containing requested object, and sends message into its socket

time



Non-persistent HTTP: example (cont.)

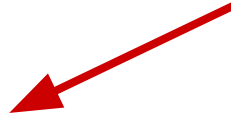
User enters URL: `www.someSchool.edu/someDepartment/home.index`
(containing text, references to 10 jpeg images)



5. HTTP client receives response message containing html file, displays html. Parsing html file, finds 10 referenced jpeg objects

6. Steps 1-5 repeated for each of 10 jpeg objects

4. HTTP server closes TCP connection.

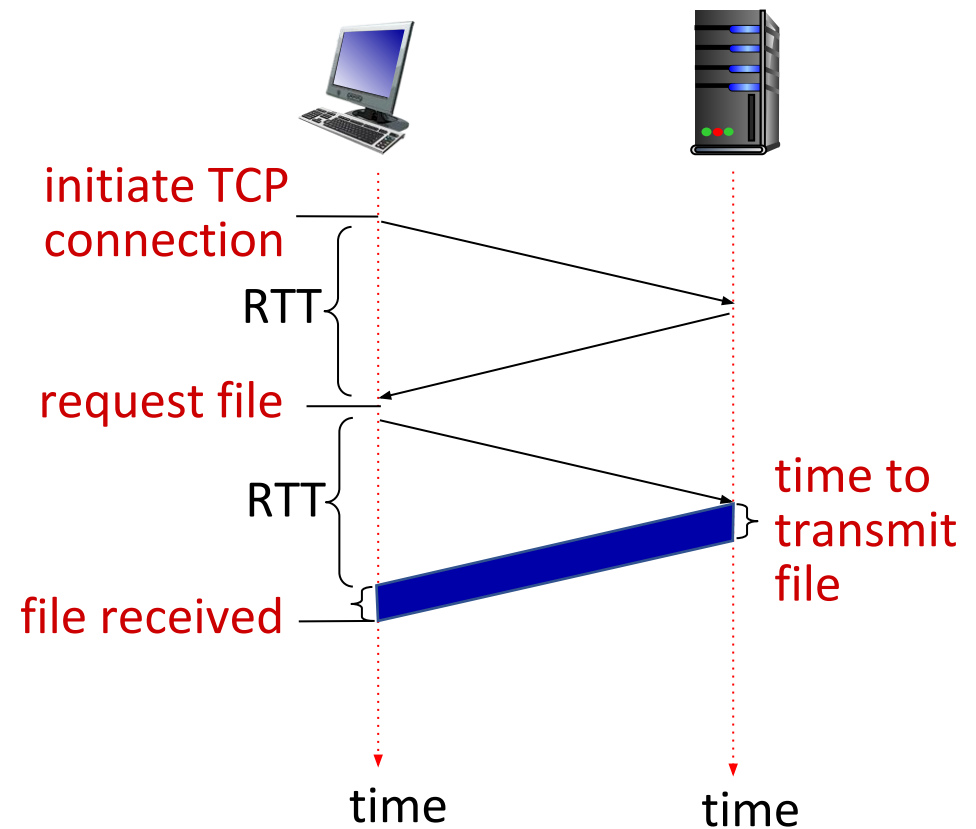


Non-persistent HTTP: response time

RTT (definition): time for a small packet to travel from client to server and back

HTTP response time (per object):

- one RTT to initiate TCP connection
- one RTT for HTTP request and first few bytes of HTTP response to return
- object/file transmission time



Non-persistent HTTP response time = $2RTT + \text{file transmission time}$

Persistent HTTP (HTTP 1.1)

Non-persistent HTTP issues:

- requires 2 RTTs per object
- OS overhead for *each* TCP connection
- browsers often open multiple parallel TCP connections to fetch referenced objects in parallel

Persistent HTTP (HTTP1.1):

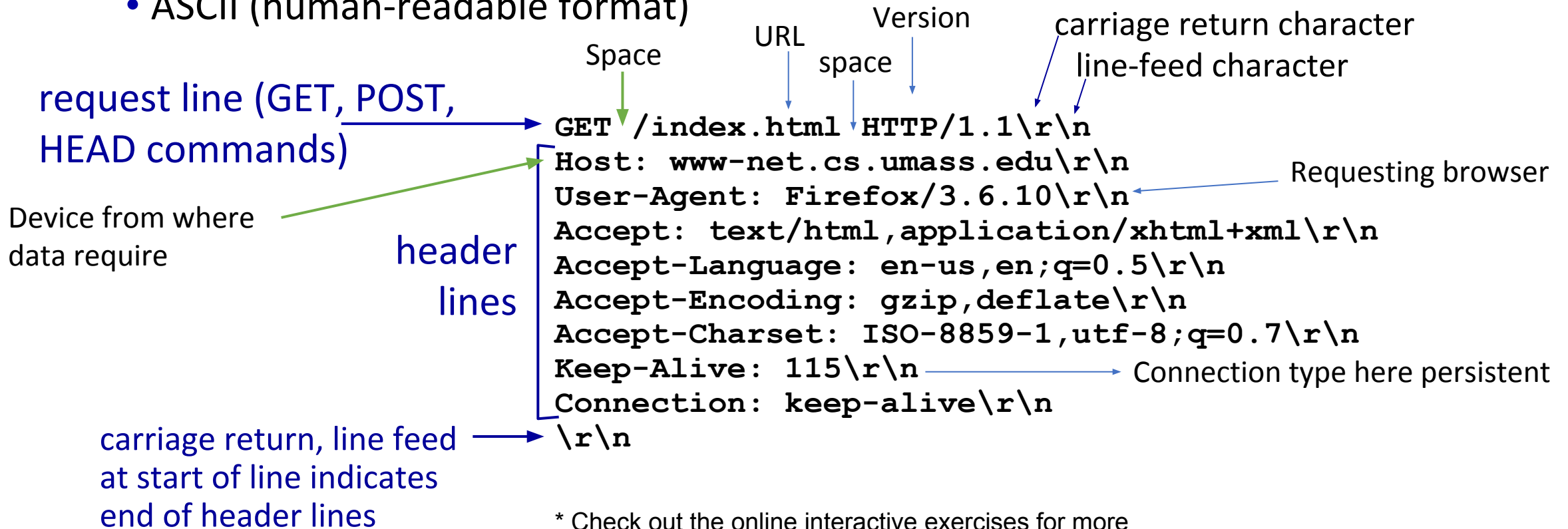
- server leaves connection open after sending response
- subsequent HTTP messages between same client/server sent over open connection
- client sends requests as soon as it encounters a referenced object
- as little as one RTT for all the referenced objects (cutting response time in half)

HTTP request message

- two types of HTTP messages: *request, response*

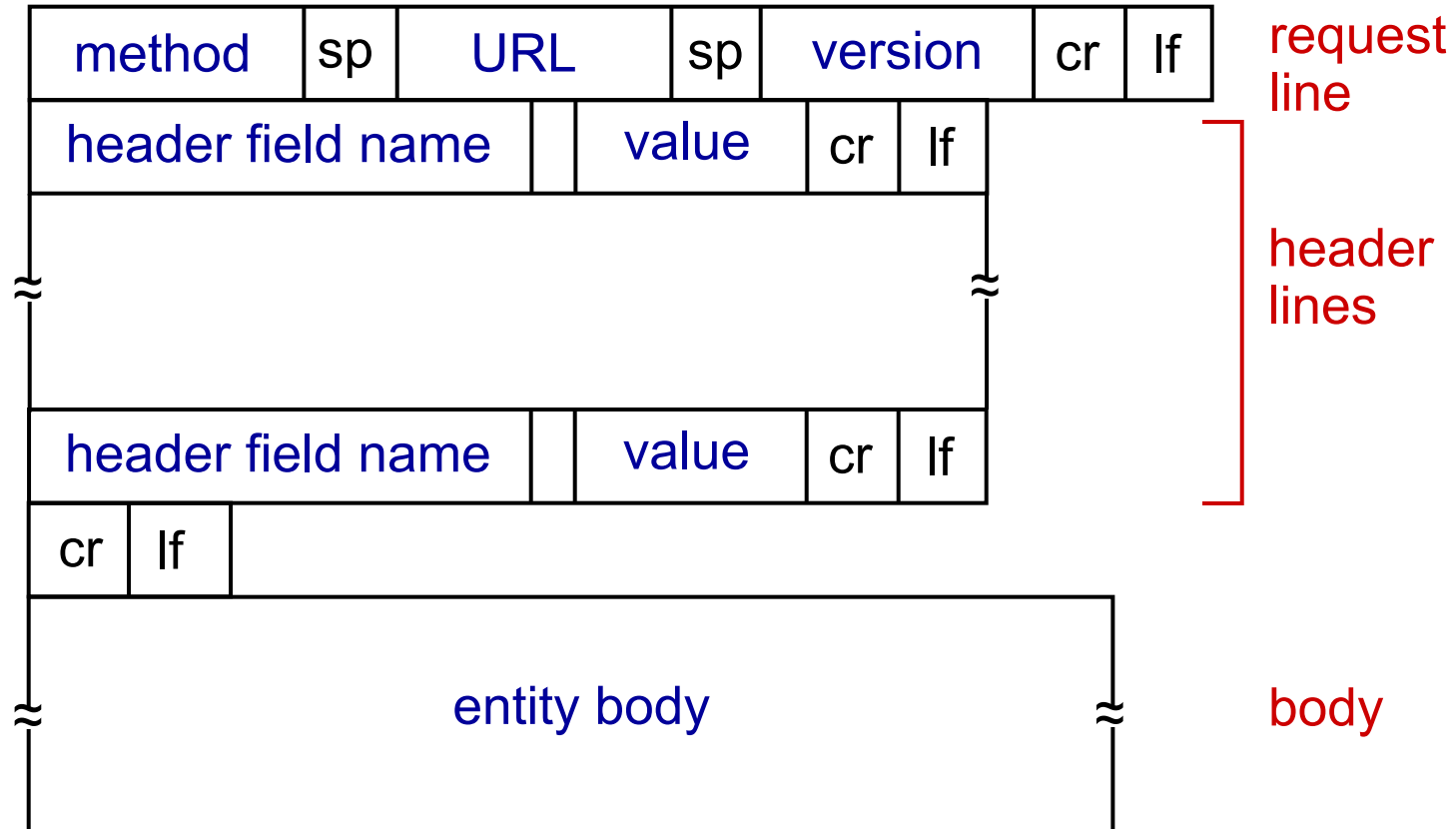
- HTTP request message:

- ASCII (human-readable format)



* Check out the online interactive exercises for more examples: http://gaia.cs.umass.edu/kurose_ross/interactive/

HTTP request message: general format



Other HTTP request messages

POST method:

- web page often includes form input
- user input sent from client to server in entity body of HTTP POST request message

GET method (for sending data to server):

- include user data in URL field of HTTP GET request message (following a '?'):

`www.somesite.com/animalsearch?monkeys&banana`

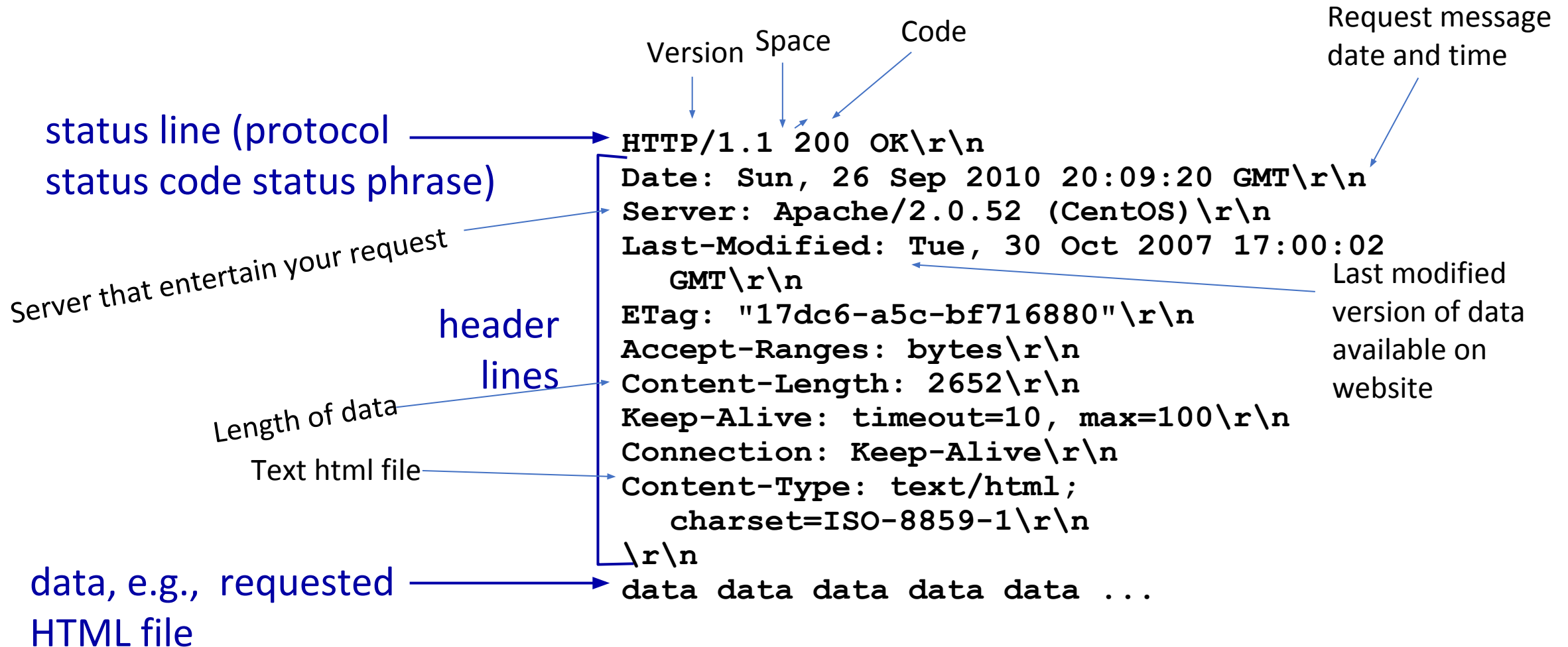
HEAD method:

- requests headers (only) that would be returned *if* specified URL were requested with an HTTP GET method.

PUT method:

- uploads new file (object) to server
- completely replaces file that exists at specified URL with content in entity body of POST HTTP request message

HTTP response message



* Check out the online interactive exercises for more examples: http://gaia.cs.umass.edu/kurose_ross/interactive/

HTTP response status codes

- status code appears in 1st line in server-to-client response message.
- some sample codes:

200 OK

- request succeeded, requested object later in this message

301 Moved Permanently

- requested object moved, new location specified later in this message (in Location: field)

400 Bad Request

- request msg not understood by server

404 Not Found

- requested document not found on this server

505 HTTP Version Not Supported

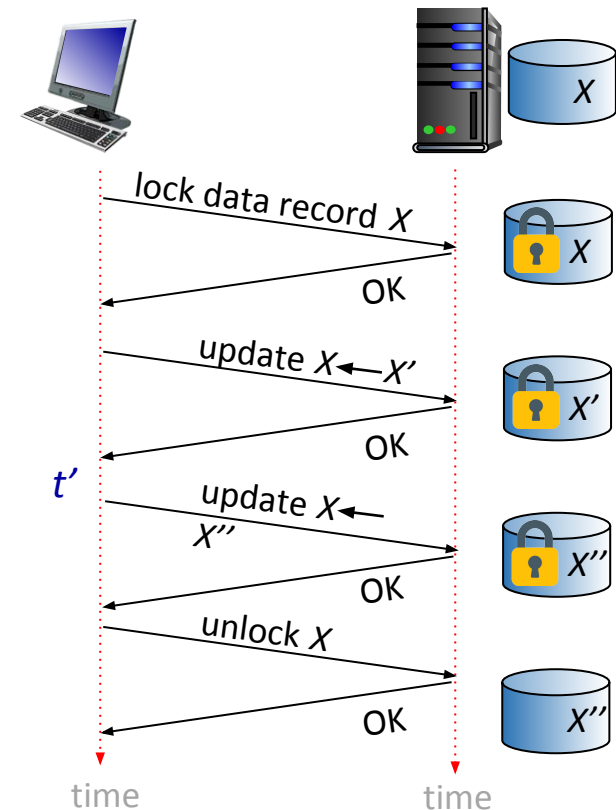
When browser and server http versions not compatible

Maintaining user/server state: cookies

Recall: HTTP GET/response interaction is *stateless*

- no notion of multi-step exchanges of HTTP messages to complete a Web “transaction”
 - no need for client/server to track “state” of multi-step exchange
 - all HTTP requests are independent of each other
 - no need for client/server to “recover” from a partially-completed-but-never-completely-completed transaction

a *stateful* protocol: client makes two changes to X, or none at all



Q: what happens if network connection or client crashes at t' ?

Maintaining user/server state: cookies

Web sites and client browser use *cookies* to maintain some state between transactions

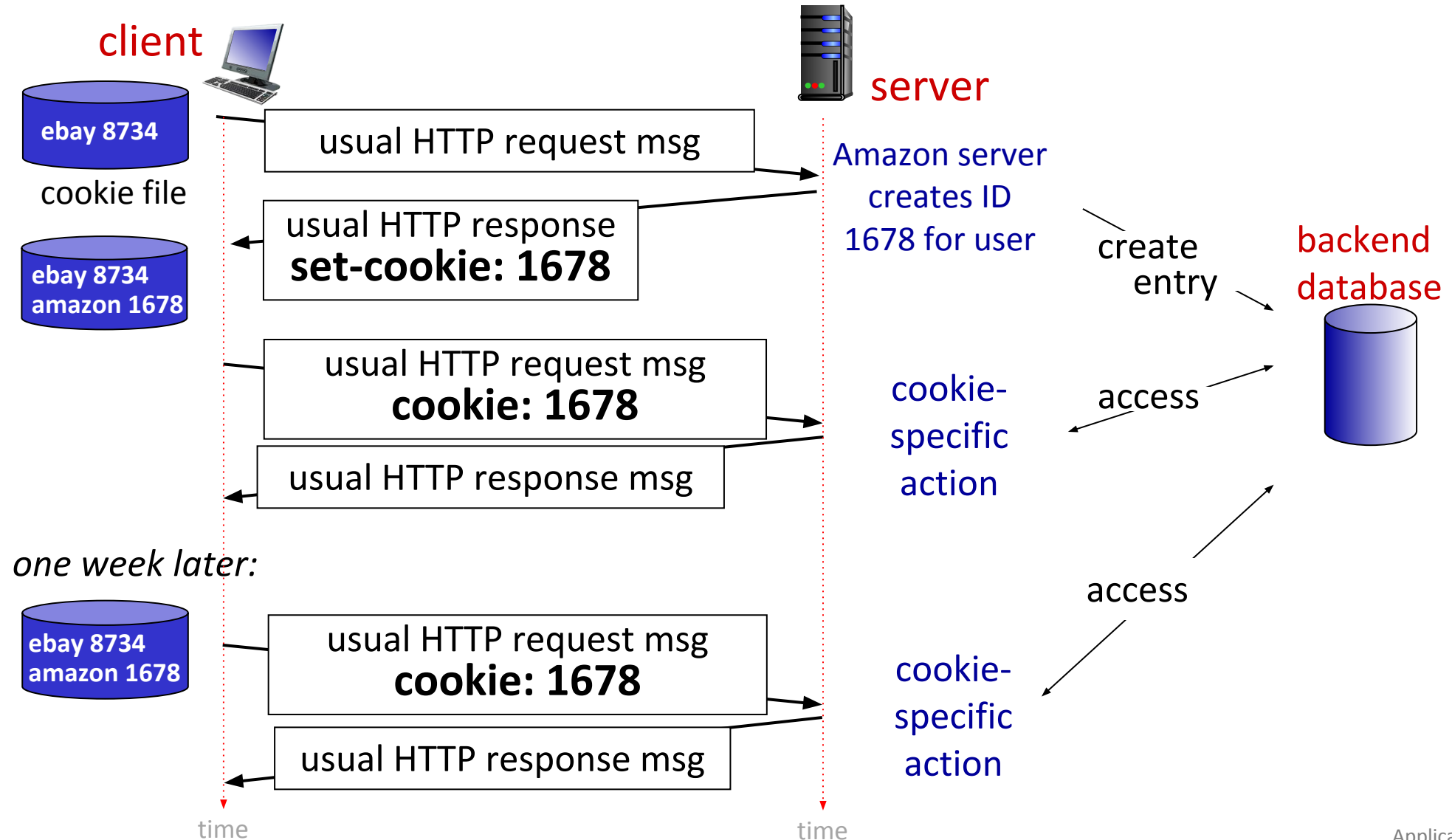
four components:

- 1) cookie header line of HTTP *response* message
- 2) cookie header line in next HTTP *request* message
- 3) cookie file kept on user's host, managed by user's browser
- 4) back-end database at Web site

Example:

- Susan uses browser on laptop, visits specific e-commerce site for first time
- when initial HTTP requests arrives at site, site creates:
 - unique ID (aka “cookie”)
 - entry in backend database for ID
- subsequent HTTP requests from Susan to this site will contain cookie ID value, allowing site to “identify” Susan

Maintaining user/server state: cookies



HTTP cookies: comments

What cookies can be used for:

- authorization
- shopping carts
- recommendations
- user session state (Web e-mail)

Challenge: How to keep state:

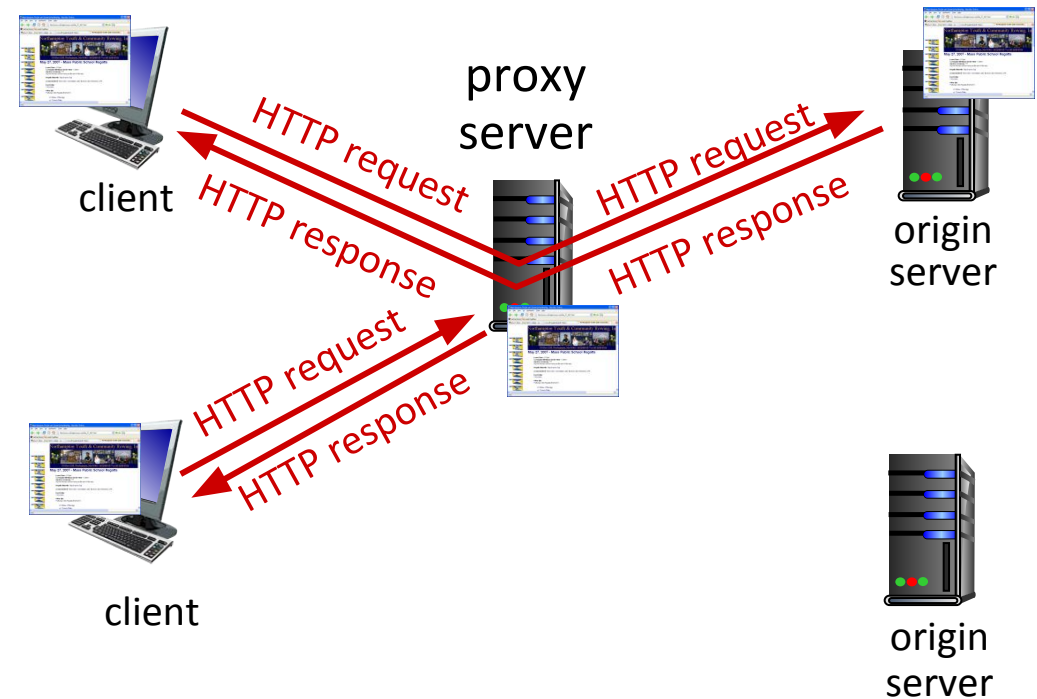
- protocol endpoints means sender and receiver: maintain state at sender side browser and at receiver side browser over multiple transactions
- In cookies: HTTP messages carry state

- aside
- cookies and privacy:*
- cookies permit sites to *learn* a lot about you on their site.
 - third party persistent cookies (tracking cookies) allow common identity (cookie value) to be tracked across multiple web sites

Web caches (proxy servers)

Goal: satisfy client request without involving origin server

- user configures browser to point to a *Web cache*
- browser sends all HTTP requests to cache
 - *if* object in cache: cache returns object to client
 - *else* cache requests object from origin server, caches received object, then returns object to client



Web caches (proxy servers)

- Web cache acts as both client and server
 - server for original requesting client
 - client to origin server
- typically cache is installed by ISP (university, company, residential ISP)

Why Web caching?

- reduce response time for client request
 - cache is closer to client
- reduce traffic on an institution's access link
- Internet is dense with caches
 - enables “poor” content providers to more effectively deliver content

HTTP/2

Key goal: decreased delay in multi-object HTTP requests

HTTP1.1: introduced **multiple, pipelined GETs** over single TCP connection

- server responds *in-order* (FCFS: first-come-first-served scheduling) to GET requests
- **HOL = Head Of Line** means the FIRST response blocks everything behind it.
- **Example :** Request 1 = large video file, Request 2 = small icon (only 1 KB), Request 3 = small CSS file
- **Server must send:** → huge → tiny → tiny The small responses (2 & 3) must wait until the large one finishes
- loss recovery (retransmitting lost TCP segments) stalls object transmission

HTTP/2

Key goal: decreased delay in multi-object HTTP requests

HTTP/2: [RFC 7540, 2015] increased flexibility at *server* in sending objects to client:

- methods, status codes, most header fields unchanged from HTTP 1.1
- transmission order of requested objects based on client-specified object priority (not necessarily FCFS)
- **Example** :Browser may say: CSS file → **highest priority** JavaScript → **medium priority** large images → **low priority**
- *Server can send objects that client did not request yet, but will likely need.*
- divide objects into frames, schedule frames to mitigate HOL blocking

HTTP/2 to HTTP/3

Key goal: decreased delay in multi-object HTTP requests

HTTP/2 over single TCP connection means:

- recovery from packet loss still stalls all object transmissions
 - as in HTTP 1.1, browsers have incentive to open multiple parallel TCP connections to reduce stalling, increase overall throughput
 - In short **1 TCP connection lost a packet → all streams pause**
- no security over vanilla TCP connection(transport layer provide security not protocol)
- **HTTP/3:** adds security , per object error- and congestion-control (more pipelining) over UDP
 - more on HTTP/3 in transport layer

Application layer: overview

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- Web and HTTP
- E-mail, SMTP, IMAP
- The Domain Name System DNS
- P2P applications
- video streaming and content distribution networks
- socket programming with UDP and TCP



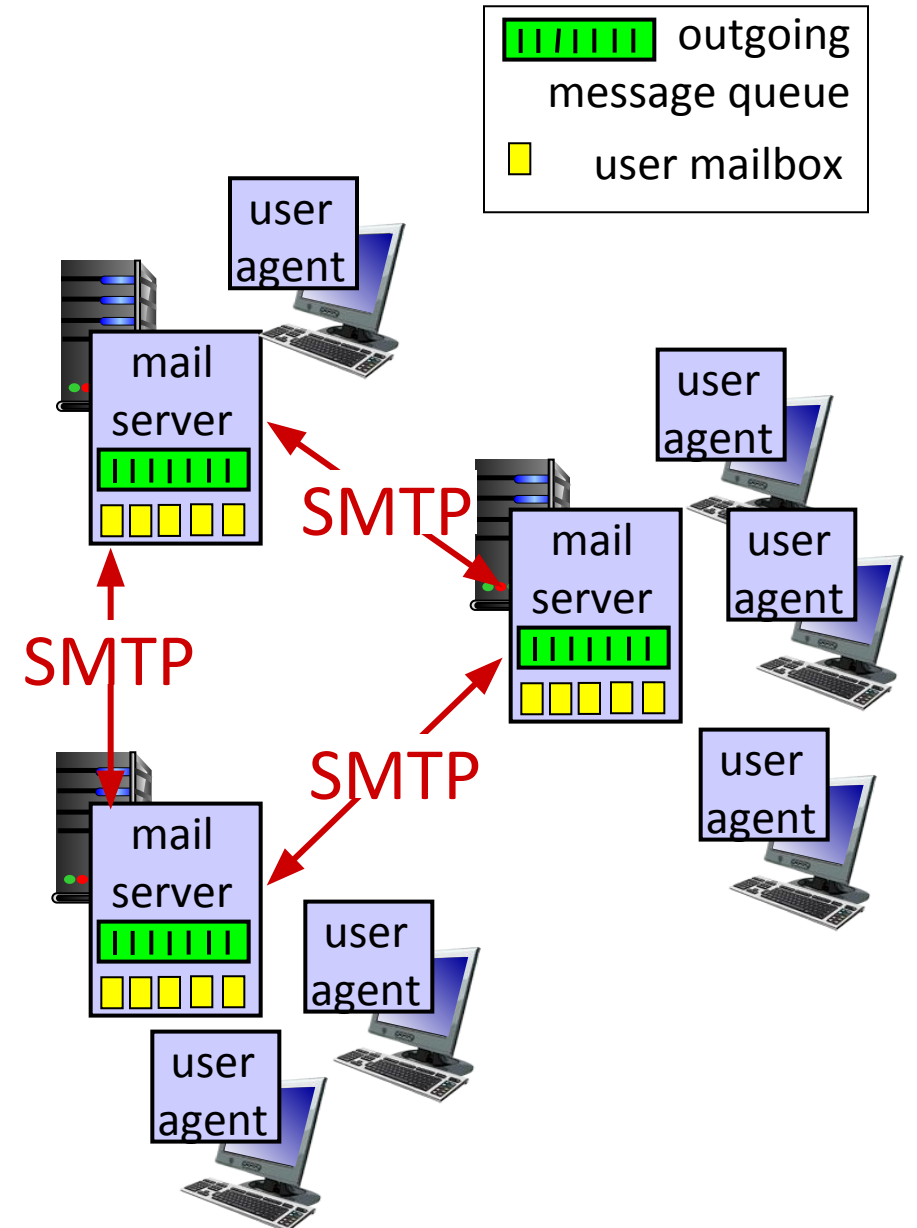
E-mail

Three major components:

- user agents
- mail servers
- simple mail transfer protocol: SMTP

User Agent

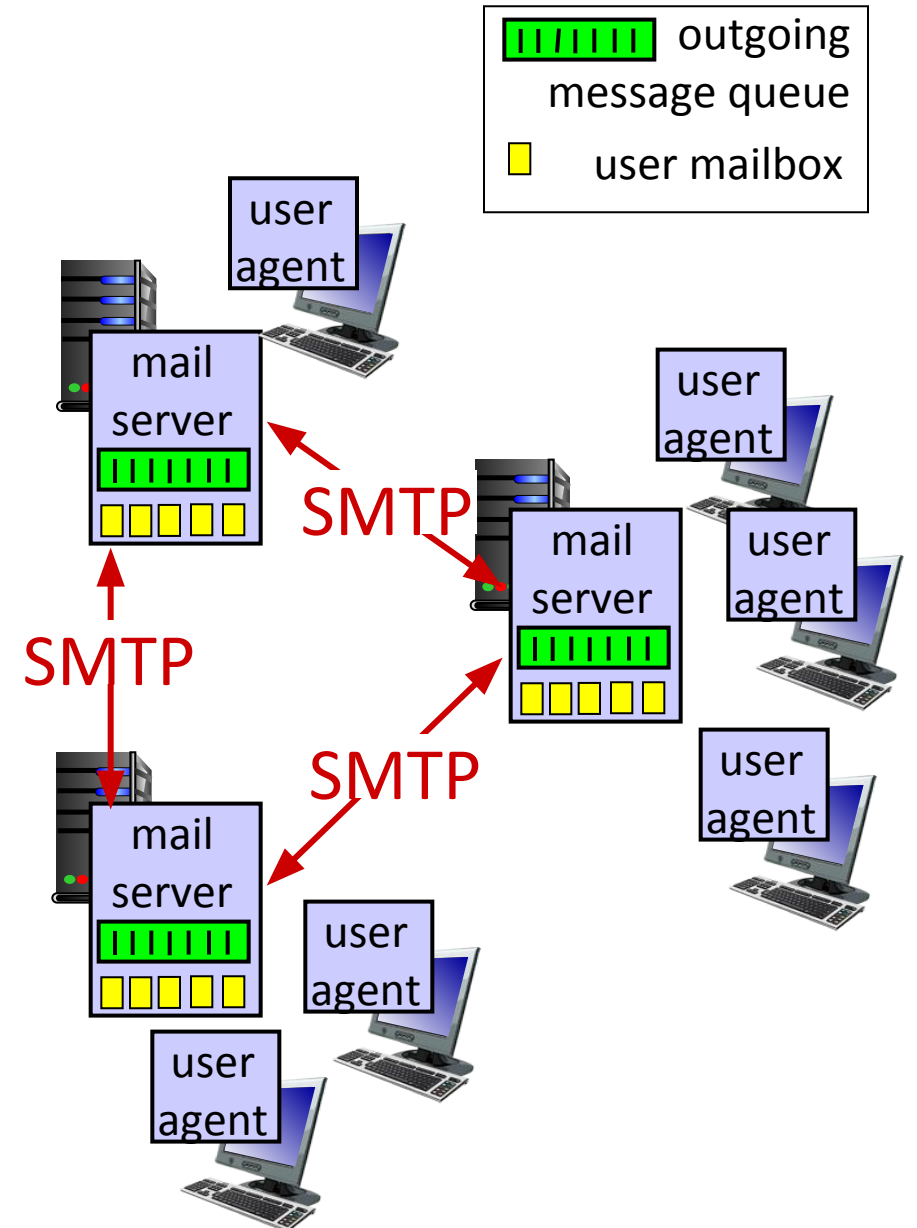
- a.k.a. “mail reader”
- composing, editing, reading mail messages
- e.g., Outlook, iPhone mail client
- outgoing, incoming messages stored on server



E-mail: mail servers

mail servers:

- *mailbox* contains incoming messages for user
- *message queue* of outgoing (to be sent) mail messages
- *SMTP protocol* between mail servers to send email messages
 - client: sending mail server
 - “server”: receiving mail server

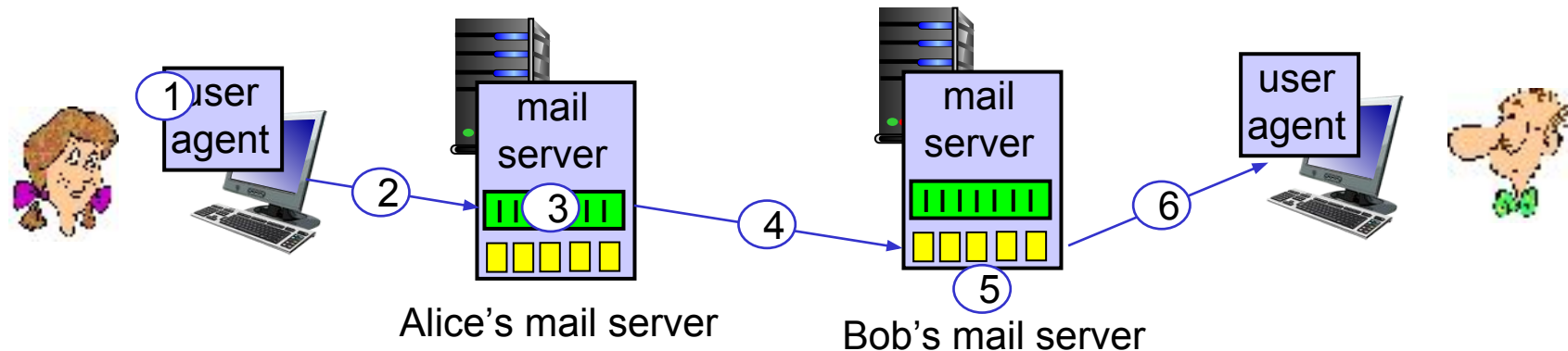


E-mail: the RFC (5321)

- uses TCP to reliably transfer email message from client (mail server initiating connection) to server, port 25
- **direct transfer:** sending server (acting like client) to receiving server
- three phases of transfer
 - handshaking (greeting)
 - transfer of messages
 - closure
- command/response interaction (like HTTP)
 - **commands:** ASCII text i.e. **HELLO mymail.com**
 - **response:** status code and phrase i.e. **250 Hello**
- messages must be in 7-bit ASCII

Scenario: Alice sends e-mail to Bob

- 1) Alice uses UA to compose e-mail message "to" bob@some school.edu
- 2) Alice's UA sends message to her mail server; message placed in message queue
- 3) client side of SMTP opens TCP connection with Bob's mail server
- 4) SMTP client sends Alice's message over the TCP connection
- 5) Bob's mail server places the message in Bob's mailbox
- 6) Bob invokes his user agent to read message



Sample SMTP interaction

S=Server
C= client

```
S: 220 hamburger.edu //220 = status code meaning "Service ready"
C: HELO crepes.fr
S: 250 Hello crepes.fr, pleased to meet you //Server responds 250 = OK
C: MAIL FROM: <alice@crepes.fr> //Client tells server who is sending the email.
S: 250 alice@crepes.fr... Sender ok // Server responds 250 = OK = Sender accepted
C: RCPT TO: <bob@hamburger.edu> //Client tells server who the email is for
S: 250 bob@hamburger.edu ... Recipient ok
C: DATA
S: 354 Enter mail, end with "." on a line by itself 354 → ready to accept message
C: Do you like ketchup? //email text lines
C: How about pickles?
C: . //indicate end of email
S: 250 Message accepted for delivery
C: QUIT
S: 221 hamburger.edu closing connection
```

SMTP: closing observations

comparison with HTTP:

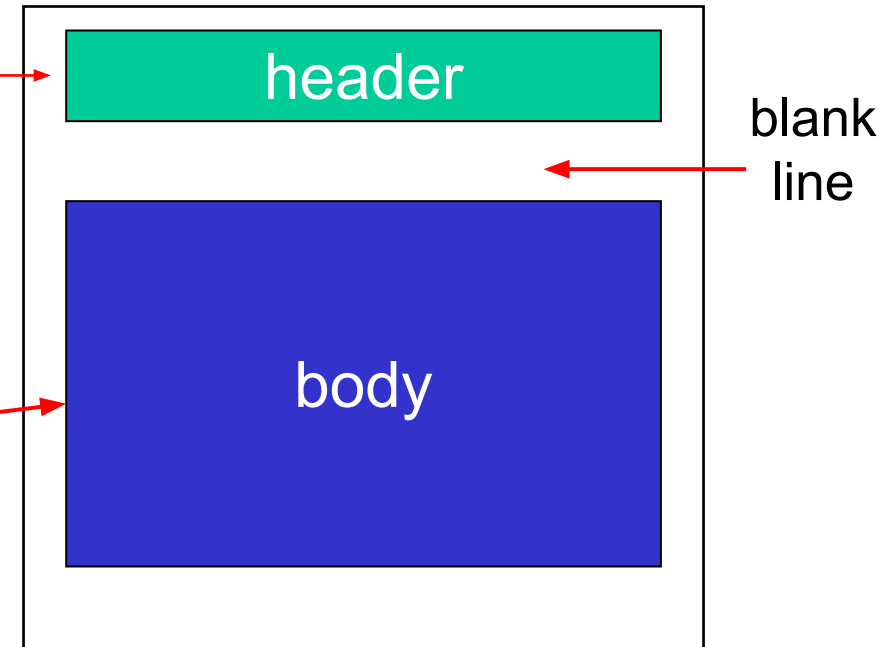
- HTTP: **pull**(Client requests (pulls) each resource from server)
- SMTP: **push**(mail Server pushes email to recipient mail server)
- both have ASCII command/response interaction, status codes
- **HTTP**: Pull-based, 1 object per response, ASCII commands, status codes
- **SMTP**: Push-based, multipart messages, ASCII commands, status codes
- SMTP uses persistent connections
- SMTP requires message (header & body) to be in 7-bit ASCII
- SMTP server uses CRLF.CRLF to determine end of message
- CRLF = Carriage Return + Line Feed (new line)Single dot . on a line by itself signals end of DATA command.

Mail message format

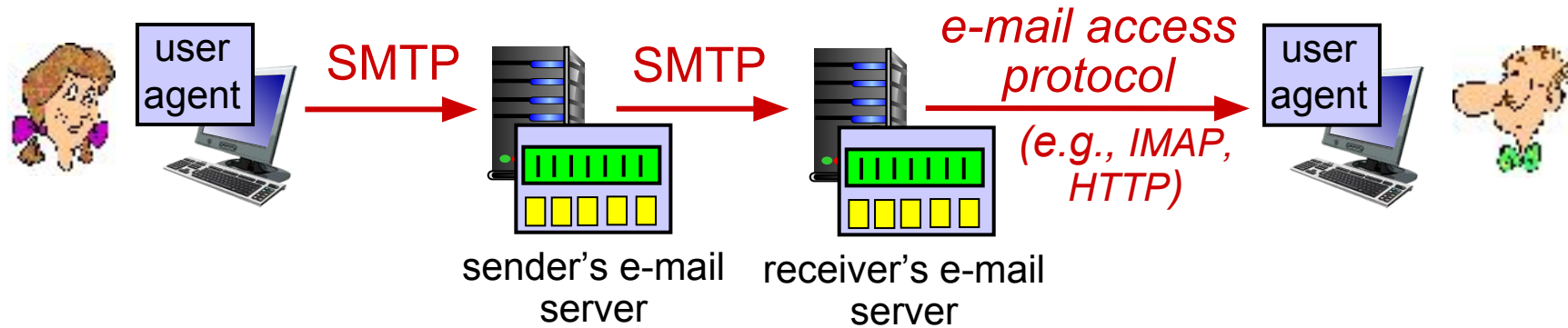
SMTP: protocol for exchanging e-mail messages, defined in RFC 531 (like HTTP)

RFC 822 defines *syntax* for e-mail message itself (like HTML)

- header lines, e.g.,
 - To:
 - From:
 - Subject:these lines, within the body of the email message area different from SMTP MAIL FROM., RCPT TO: commands!
- Body: the “message” , ASCII characters only



Mail access protocols



- **SMTP**: delivery/storage of e-mail messages to receiver's server
- mail access protocol: retrieval from server
 - **IMAP**: Internet Mail Access Protocol [RFC 3501]: messages stored on server, IMAP provides retrieval, deletion, folders of stored messages on server
- **HTTP**: gmail, Hotmail, Yahoo!Mail, etc. provides web-based interface on top of SMTP (to send), IMAP (or POP) to retrieve e-mail messages